Ryan Diaz

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Education

Rice University — Aug. 2025 – Present
Doctor of Philosophy, Computer Science — *Houston, TX*

University of Minnesota, Twin Cities — Sep. 2021 – May 2025
Bachelor of Science, Computer Science and Mathematics — *Minneapolis, MN*

- **Honors:** summa cum laude with high distinction (GPA 4.0/4.0)
- **Relevant Coursework:** Deep Learning for Robot Manipulation, Machine Learning, Data Analysis, Computer Vision, Natural Language Processing, Stochastic Processes, Artificial Intelligence, Linear Algebra, Probability and Statistics

Research Experience

Human-AI Teaming Lab, Rice University — Aug. 2025 – Present
Graduate Researcher, Advised by Prof. Vaibhav Unhelkar — *Houston, TX*

Robotics: Perception and Manipulation Lab, University of Minnesota, Twin Cities — Dec. 2022 – May 2025
Undergraduate Researcher, Advised by Prof. Karthik Desingh — *Minneapolis, MN*

CERL Lab, Washington University in St. Louis — May 2024 – Aug. 2024
Undergraduate Researcher (NSF REU), Advised by Prof. Yevgeniy Vorobeychik — *St. Louis, MO*

Awards and Honors

NSF Graduate Research Fellowship Program (GRFP) Award	Apr. 2026
IROS Student and Developing Countries (SDC) Travel Award	Aug. 2025
UMN College of Science and Engineering Dean's List (x8)	Dec. 2021 – May 2025
CRA Outstanding Undergraduate Researcher Award (Honorable Mention)	Jan. 2025
UMN Undergraduate Research Opportunities Program (UROP) Award (x2)	Aug. 2023 – Aug. 2024
UMN Prof. Hans H. Dalaker Mathematics Scholarship Award	Jun. 2024
UMN Hopper-Dean Foundation Computer Science Scholarship Award	Jun. 2024
UMN Ella Thorp Mathematics Scholarship Award	May 2023
UMN Undergraduate Research Scholarship (URS) Award	Jan. 2023
UMN Gold Scholar Award	Sep. 2021

Publications

Conference Publications

C1. Zhiqin Qian, **Ryan Diaz**, Sangwon Seo, and Vaibhav Unhelkar, "Hierarchical Reward Design from Language: Enhancing Alignment of Agent Behavior with Human Specifications," in *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2026.

C2. **Ryan Diaz**, Adam Imdieke, Vivek Veeriah, and Karthik Desingh, "AugInsert: Learning Robust Visual-Force Policies via Data Augmentation for Object Assembly Tasks," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

C3. Chahyon Ku, Carl Winge, **Ryan Diaz**, Wentao Yuan, and Karthik Desingh, "Evaluating Robustness of Visual Representations for Object Assembly Task Requiring Spatio-Geometrical Reasoning," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

Peer-Reviewed Workshop Papers

W1. **Ryan Diaz** and Vaibhav Unhelkar, "CO-OPT: Lifelong Learning of Object-Oriented Preferences for Personalized Robotic Assistance," in *Workshop on Learning from Corrections and Interventions @ Conference on Robot Learning (CoRL)*, 2026.

Presentations

Oral/Poster: IEEE/RSJ IROS 2025 — Oct. 2025
"Learning Robust Visual-Force Policies via Data Augmentation for Object Assembly Tasks" — *Hangzhou, CN*

Poster: WashU STEM Poster Palooza — Aug. 2024
"Vision-Based Algorithms for Obstacle Detection and Avoidance in Autonomous Vehicles" — *St. Louis, MO*

Video: UMN Undergraduate Research Symposium — Dec. 2023
"Augmenting a Dual-Arm Contact-Rich Robotic Manipulation Task with Force-Torque Data" — *Minneapolis, MN*

Poster: UMN Summer Undergraduate Research Expo — Aug. 2023
"Imitation Learning for Spatio-Geometry Driven Assembly Task with Dual-Arm Manipulator" — *Minneapolis, MN*

Poster: UMN Undergraduate Research Symposium — Apr. 2023
"Large-Scale Object Generation for Learning Robotic Manipulation Tasks" — *Minneapolis, MN*

Teaching Experience

COMP 459/559: Machine Learning with Graphs — Spring 2026
Graduate Teaching Assistant — *Rice University*

CSCI 4521: Applied Machine Learning — Spring 2025
Undergraduate Teaching Assistant — *University of Minnesota, Twin Cities*

MATH 5652: Introduction to Stochastic Processes — Spring 2025
Undergraduate Paper Grader — *University of Minnesota, Twin Cities*

CSCI 2033: Elementary Computational Linear Algebra — Spring 2023, Fall 2023, Fall 2024
Undergraduate Teaching Assistant — *University of Minnesota, Twin Cities*

CSCI 4511W: Introduction to Artificial Intelligence — Spring 2024
Undergraduate Teaching Assistant — *University of Minnesota, Twin Cities*

CSCI 1933: Introduction to Algorithms and Data Structures — Summer 2023
Undergraduate Teaching Assistant — *University of Minnesota, Twin Cities*

UMN Taylor Tutoring Center — Fall 2022
Undergraduate Peer Tutor — *University of Minnesota, Twin Cities*

Academic Service

Conference Reviewer
ICRA (2025), *IROS* (2025, 2026)

Workshop Reviewer
CoRL Workshop on Learning from Corrections and Interventions (2026)

Undergraduate Peer Reviewer
Minnesota Undergraduate Research and Academic Journal (2024, 2025)

Mentoring

Terence Li, *Undergraduate @ Rice University* — Oct. 2026 – Present
Project: Investigating Human Communication Strategies in Long-Term HRI

Gael Cisneros, *High School Student* — Jun. 2026 – Aug. 2026
Project: Facilitating More Natural HRI with a Robot Face

Shubham Panchal, *High School Student* — Jan. 2026 – May 2026
Project: Few-Shot Manipulation via Task-Aware Model-Agnostic Meta-Learning

Technical Skills

Languages: Python, Java, C, C++, JavaScript, HTML, MATLAB, Bash, LaTeX
Libraries: PyTorch, Tensorflow, OpenCV, NumPy, ROS(2), Transformers (HuggingFace), MuJoCo
Software: Git, Linux, Blender